

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

Mid-Semester Examination, Spring 2023-24

B. Tech. 6th Semester

Subject Name: Computer Networks

Subject Code: CS3002

Number of pages: 1

Maximum Marks: 30

Exam Duration: 2 Hours

Note: There are Five questions. Attempt All the questions. All parts of the same question must be attempted in a sequence.

Q.1 (a) Differentiate between *packet switching* and *circuit switching*.

(b) List out five network topologies. Which among those listed is least prone to failure? Explain your reasoning.

(c) Differentiate between store-and-forward switching and cut-through switching.

(d) Discuss any two functionalities of the Transport layer and Session layer of the OSI Model.

(e) What do you understand by HTTP? Differentiate between Non-persistent and Persistent HTTP. Also, compare their response time in terms of RTT and transmission time.

[2x5=10 Marks]

Q.2 Consider a scenario with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 300$ Mbps. The four links (each) from the servers to the shared link have a transmission capacity of $R_S = 30$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 90$ Mbps.

(i) What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

(ii) Which link is the bottleneck link? Format as R_C , R_S , or R .

(iii) Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S) and client links (R_C)?

[1+1+2=4 Marks]

Q.3 State whether true or false and also justify your answer:

(a) HTTP is in-band and FTP is out-band **(b)** Packet switching is a connection-oriented service

(c) HTTP is a state property **(d)** SMTP is a pull protocol

[1.5X4=6 Marks]

Q.4 How are DNS requests and replies carried out? What is Resource Records(RR) in DNS protocol? Discuss the four types of RR in detail with the help of examples.

[1+1+2=4 Marks]

Q.5 A XYZ company has 50 Mbps of LAN connectivity and 10Mbps internet bandwidth connected via an access link. Assume the average request rate from the company's browsers to origin servers = λ requests/sec and delay from company access router to any origin server and back to router = 3 sec. Consider $\lambda=5$ and an object of size 1000000 bits. Calculate the following:

(a.) Utilization of LAN and access link

(b.) Total delay from host of XYZ Network to the origin server to get the response.

(c.) Assume the hit ratio =0.5, calculate the total delay of part (b)

(d.) Also calculate the percentage of reduction in total delay due to web caching of part(c).

[1+2+2+1=6 Marks]